



Phylogenomics reveals extensive introgression and a case of mito-nuclear discordance in the killifish genus *Kryptolebias*

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ABSTRACT

Introgression is a widespread evolutionary process leading to phylogenetic inconsistencies among distinct parts of the genomes, particularly between mitochondrial and nuclear-based phylogenetic reconstructions (e.g., mito-nuclear discordances). Here, we used mtDNA and genome-wide nuclear sites to provide the first phylogenomic-based hypothesis on the evolutionary relationships within the killifish genus *Kryptolebias*. In addition, we tested for evidence of past introgression in the genus given the multiple reports of undergoing hybridization between its members. Our mtDNA phylogeny generally agreed with the relationships previously proposed for the genus. However, our reconstruction based on nuclear DNA revealed an unknown lineage - *Kryptolebias* sp. 'ESP' - as the sister group of the self-fertilizing mangrove killifishes, *K. marmoratus* and *K. hermaphroditus*. All individuals sequenced of *Kryptolebias* sp. 'ESP' had the same mtDNA haplotype commonly observed in *K. hermaphroditus*, demonstrating a clear case of mito-nuclear discordance. Our analysis further confirmed extensive history of introgression between *Kryptolebias* sp. 'ESP' and *K. hermaphroditus*. Population genomics analyses indicate no current gene flow between the two lineages, despite their current sympatry and history of introgression. We also confirmed introgression between other species pairs in the genus that have been recently reported to form hybrid zones. Overall, our study provides a phylogenomic reconstruction covering most of the *Kryptolebias* species, reveals a new lineage hidden in a case of mito-nuclear discordance, and provides evidence of multiple events of ancestral introgression in the genus. These findings underscore the importance of investigating different genomic information in a phylogenetic framework, particularly in taxa where introgression is common as in the sexually diverse mangrove killifishes.

1. Introduction

Estimating the evolutionary relationships among species is a crucial goal of evolutionary biology. With the unprecedented availability of large numbers of loci brought by the genomics era, it has become increasingly clear that organisms generally have a more complex evolutionary history than previously acknowledged, with biological processes such as recombination, incomplete lineage sorting, introgression, and genome rearrangements (Mallet et al., 2016; Nakhleh, 2013) contributing to different phylogenetic signals among topologies

generated from different sets of data for the same group of organisms (Bravo et al., 2019).

Although phylogenetic incongruence may have appeared as a problem at first (Jeffroy et al., 2006; Maddison, 1997), evolutionary biologists now embrace heterogeneity of phylogenetic signals (Bravo et al., 2019; Hahn and Nakhleh, 2016), recognizing that phylogenetic incongruences offer a unique opportunity to investigate the biological phenomena underlying discordance. Among these, reticulate evolution through introgression is today commonly accepted as a widespread evolutionary process contributing to phylogenetic discordance (Bravo

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